

Jiajie(Jason) Ma

Duke University
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Education

Duke University	2022 – Present
Ph.D in Mathematics (Expected May 2027), Certificate in College Teaching Advisor: Lenhard Ng	
Haverford College	2018 – 2022
B.S. in Mathematics and Computer Science, <i>summa cum laude</i> , <i>Phi Beta Kappa</i>	
Budapest Semester in Mathematics	Fall 2020

Visiting positions

IPAM, UCLA	Fall 2026
Graduate fellow, long program “quantum topology, character varieties and low-dimensional geometry”	
RIKEN, Sendai	Summer 2024
Visiting graduate student, SLMath summer graduate school “h-principle”	

Research Interests

Symplectic geometry, contact geometry, and low-dimensional topology

Honors and Awards

L.P. and Barbara Smith Award for Excellence in Teaching	2026
<i>Duke University</i>	
L.P. and Barbara Smith Award for Promise of Teaching Excellence	2025
<i>Duke University</i>	

Publications

7. **Functoriality of the L_∞ structure for Legendrian contact homology,**
in preparation.
6. **Cluster structures from Legendrian double twist knots: a comparative approach,**
joint with James Hughes, arXiv:2610.02086 (2026).
5. **Relative Calabi-Yau Structures for Legendrian Contact Homology,**
joint with Joshua M. Sabloff, arXiv:2509.02485 (2025).

4. **Exact Lagrangian fillability of 3-braid closures**,
joint with James Hughes, *Bulletin of London Mathematical Society* 58-1 (2026), 1-20
3. **On the monodromy and spin parity of single-cylinder origamis in the minimal stratum**,
joint with Tarik Aougab, Adam Friedman-Brown, and Luke Jeffreys, arXiv:2502.09498 (2025).
2. **On faces of the Kunz cone and the numerical semigroups within them**,
joint with Levi Borevitz, Tara Gomes, Harper Niergarth, Christopher O'Neill, Daniel Pocklington, Rosa Stolk, Jessica Wang, and Shuhang Xue, to appear in *Rocky Mountain Journal of Mathematics* (2023).
1. **Information access representations and social capital in networks**,
joint with Ashkan Bashardoust, Hannah C. Beilinson, Sorelle A. Friedler, Jade Rousseau, Carlos E. Scheidegger, Blair D. Sullivan, Nasanbayar Ulzii-Orshikh, and Suresh Venkatasubramanian, arXiv:2010.12611 (2023).

Research Talks

AMS Eastern Sectional Meeting, Sheaf-Theoretic and Floer-Theoretic Approaches to Contact and Symplectic Topology
 Functoriality of the L_∞ structure for Legendrian contact homology March 2026

Conferences and Workshops

CBMS Summer School: Legendrian Links & Microlocal Sheaves June 2025
 Georgia Topology Summer School: knotted surfaces in four dimensions May 2024
 Erdős Center Winter School: Singularities & Low-Dimensional Topology Jan 2023
 SYNC Summer School: Symplectic & Contact Topology August 2022

Teaching Experience

At Duke University:

Instructor, MATH111L: Laboratory Calculus I Spring 2026
 Instructor, MATH111L: Laboratory Calculus I Spring 2025
 Discussion Leader, MATH218D: Matrices and Vector Spaces Fall 2024
 Lecture TA, MATH221D: Linear Algebra, lecture TA Spring 2024
 Lab TA, MATH111L: Laboratory Calculus I Fall 2023
 Lab TA, MATH111L: Laboratory Calculus I Fall 2022

At Haverford College:

Tutor, Math@theWritingCenter 2019 – 2022
 Tutor, Math Question Center 2019 – 2022

Service and Outreach

Project manager for 3 undergraduate researchers in Duke Math+ program <i>Project report: Symmetries of Legendrian surfaces</i>	Summer 2025
Co-organizer of Triangle Area Graduate Mathematics Conference	Spring 2025
Officer of Duke American Mathematical Society (AMS) Chapter	2024 - 2025
Project manager for 4 undergraduate researchers in Duke Math+ program <i>Project report: Computing hyperbolic structures from link diagrams</i>	Summer 2023

Languages

Fluent in Mandarin and English. Elementary French and Japanese.